Today

- Axiomatic principles of quantum mechanics
- Dirac's bra-ket notation
- Position and momentum representations
- General uncertainty principle

3 Principles of quantum mechanics

Now we organize the principles of quantum mechanics more formally and axiomatically, and investigate their consequences. At the same time, we introduce and use the Dirac bra-ket notation that becomes convenient as we go along.

3.1 Axiomatic principles

1. **State vector axiom:** The state of a physical system is represented by a normalized state vector in Hilbert space. The state vector contains all information about the system. [We will represent the state vector at time t by the Dirac ket notation $|\psi(t)\rangle$, and we call it ket $\psi(t)$. We will often simply write the state as $|\psi\rangle$ with the time dependence implicitly understood.]
2. **Probability axiom:** Given a system in state $|\psi\rangle$, a measurement will find it in state $|\phi\rangle$ with probability amplitude $\langle\phi|\psi\rangle$. [We call $\langle\psi|$ a bra state, and call it bra ψ.]
3. **Hermitian operator axiom:** A physical observable (or any measurable physical quantity) is represented by a linear and Hermitian operator.
4. **Measurement axiom:**
   (a) A measurement of a physical observable results in an eigenvalue of the observable. [Denoting an observable by $\hat{A}$, we have $\hat{A}|a\rangle = a|a\rangle$, where a is the eigenvalue of $\hat{A}$ and $|a\rangle$ is the corresponding eigenvector.]
   (b) A measurement of the physical quantity represented by $\hat{A}$ collapses the state $|\psi\rangle$ before measurement into an eigenstate $|a\rangle$ of $\hat{A}$.

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5. **Time evolution axiom:** The state vector evolves according to the Schrödinger equation, $i\hbar\frac{\partial}{\partial t}|\psi(t)\rangle = \hat{H}|\psi(t)\rangle$, where $\hat{H}$ is the Hamiltonian operator for total energy.

3.2 Vector space

The state of a system is represented by a (complex) vector $|\psi\rangle$ in Hilbert space. Let us use the simple Harmonic oscillator as an example for illustration. When we studied the simple harmonic oscillator, we wrote the Schrödinger equation as $i\hbar\frac{\partial}{\partial t}\psi(x,t) = \hat{H}\psi(x,t)$, where $\hat{H} = -\frac{\hbar^2}{2m}\frac{d^2}{dx} + \frac{1}{2}m\omega^2x^2$. Recall that the wavefunctions were functions of position and time, and given by $\psi_n(x,t)$ with corresponding energies E_n, where $n = 0, 1, 2, ...$ Why didn't we construct wavefunctions that are functions of momentum and time (in stead of position and time)? Indeed, it is possible to solve the Schrödinger equation of the simple Harmonic oscillator with exactly the same level of mathematics in momentum space as in position space. For some Hamiltonians, the Schrödinger equation is simpler to solve in momentum space. More generally, we do not want to be restricted to a particular space. Therefore, we write the Schrödinger equation as

$$i\hbar\frac{\partial}{\partial t}|\psi(t)\rangle = \hat{H}|\psi(t)\rangle \tag{3.2.1}$$

without a consideration of position or momentum.

In other words, the state vector is neither in position nor in momentum space. As an example, for the simple Harmonic oscillator, we have states that correspond to $n = 0$, $n = 1$, $n = 2$, ..., $n = \infty$. The Hilbert space is the space spanned by basis vectors $|0\rangle$ for the ground state, $|1\rangle$ for the ground state, etc. We can view $|0\rangle$ as a column vector of infinite number of components with 1 in the first entry and 0 for the rest. Similarly, $|1\rangle$ could be viewed as a column vector of infinite number of components with 1 in the second entry and 0 for the rest. In other words, we may view the basis vectors as

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ \vdots \\ \vdots \end{pmatrix}, \qquad |1\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ \vdots \end{pmatrix}, \qquad |n\rangle = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ 0 \\ \vdots \end{pmatrix}, \qquad \cdots, \tag{3.2.2}$$

where the 1 in $|n\rangle$ appears in the nth entry. We denote a state by the ket $|n\rangle$ that corresponds to

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the energy level $E_n = (n + \frac{1}{2})\hbar\omega$, again without any consideration of position or momentum. Because, n runs from 0 to ∞ in discrete integers, we have an infinite dimensional discrete vector space. Because the time evolution produces a complex vector, even if we start with a real one, $|n(t)\rangle$ is a complex vector.

However, the vector spaces we will investigate also involve vectors with continuous elements as in the position of a particle which involves continuous and infinite number of points on a line. We will see how we describe such vectors later as we go along.

3.2.1 Linearity

Because the Schödinger equation is linear, given two states $|\psi_1(t)\rangle$ and $|\psi_2(t)\rangle$, the linear superposition $|\psi(t)\rangle = c_1|\psi_1(t)\rangle + c_2|\psi_2(t)\rangle$, where c_1 and c_2 are complex constants, is also a solution.

Because, the state vector needs to be normalized, multiplying a state vector by an overall complex constant does not change its content. In other words, $|\psi(t)\rangle$ and $c|\psi(t)\rangle$, where c is a complex constant, are physically equivalent.

3.2.2 Properties of a vector space

The following are properties of vector spaces under addition and multiplication by complex constants.

1. Addition:
   (a) Commutativity of addition: $|\psi_1\rangle + |\psi_2\rangle = |\psi_2\rangle + |\psi_1\rangle$.
   (b) Associativity of addition: $(|\psi_1\rangle + |\psi_2\rangle) + |\psi_3\rangle = |\psi_1\rangle + (|\psi_2\rangle) + |\psi_3\rangle$.
   (c) Existence of a zero vector $|0\rangle$ which we also write as 0: There exists a vector $|0\rangle$ such that $|\psi\rangle + |0\rangle = |\psi\rangle$.
   (d) Existence of inverse $|-\psi\rangle$: There exists a vector $|-\psi\rangle$ such that $|\psi\rangle + |-\psi\rangle = |0\rangle$.
2. Multiplication by complex numbers:
   (a) Unity: Multiplying a vector by one reproduces the same vector, $1|\psi\rangle = |\psi\rangle$.
   (b) Associativity: Given two complex numbers c_1 and c_2, $(c_1 c_2)\,|\psi\rangle = c_1\,(c_2|\psi\rangle)$.
   (c) Distributivity: Given a complex number c and two vectors $|\psi_1\rangle$ and $|\psi_2\rangle$, $c\,(|\psi_2\rangle + |\psi_1\rangle)$ $c|\psi_1\rangle + c|\psi_2\rangle$.

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3.2.3 Dual vector space

Given a vector space in which vectors $c|\psi\rangle$ lives, there exist a dual vector space in which $c|\psi\rangle$ is mapped to $c^*\langle\psi|$, where c^* is the complex conjugate of c and $c^*\langle\psi|$. In other words, for every ket vector in the vector space, there is a bra vector in the dual vector space. The bra state $c^*\langle\psi|$ is obtained by transposing and complex conjugating the ket state $c|\psi\rangle$. In other words, given a vector,

$$|\psi\rangle = \begin{pmatrix} \vdots \\ \alpha \\ \vdots \end{pmatrix}, \tag{3.2.3}$$

the dual vector is

$$\langle\psi| = \begin{pmatrix} \cdots, & \alpha^*, & \cdots \end{pmatrix}. \tag{3.2.4}$$

In particular for the basis vectors written in (3.2.2), the dual basis vectors are

$$\begin{aligned} \langle 0| = \begin{pmatrix} 1, & 0, & \cdots, & \cdots, & \cdots \end{pmatrix}, \qquad \langle 1| = \begin{pmatrix} 0, & 1, & 0, & \cdots, & \cdots \end{pmatrix}, \\ \cdots, \qquad \langle n| = \begin{pmatrix} 0, & \cdots, & 1, & 0, & \cdots \end{pmatrix}, \qquad \cdots \end{aligned} \tag{3.2.5}$$

3.2.4 Inner product

A bra $\langle\phi|$ and a ket $|\psi\rangle$ produce a complex number through the inner product (or a bra-ket) $\langle\phi|\psi\rangle = c$, where c is a complex number. The inner product has the following properties.

1. $\langle\phi|\psi\rangle = \langle\psi|\phi\rangle^*$. Therefore, $\langle\psi|\psi\rangle$ is real.
2. $\langle\psi|\psi\rangle$ is positive and finite for a normalizable ket vector. Therefore, we can multiply a given ket by a constant and choose $\langle\psi|\psi\rangle = 1$. We have $\langle\phi|\psi\rangle = 0$ for orthogonal kets $|\psi\rangle$ and $|\phi\rangle$. For an orthonormal set of kets, $\langle\psi_m|\psi_n\rangle = \delta_{mn}$.

3.3 Operators

An operator $\hat{A}$ acting on a state vector $|\psi\rangle$ transforms it into another state vector $|\phi\rangle$,

$$\hat{A}|\psi\rangle = |\phi\rangle. \tag{3.3.1}$$

Question 3.3.1: Given that $|\psi\rangle$ is a vector in Hilbert space, $\hat{A}$ must be (A). a vector, (B). a matrix, (C). a number.

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Linear operator: A linear operator acting on a linear superposition of state vectors results in a sum with the operator acting on each term individually,

$$\hat{A}\left(c_1|\psi_1\rangle + c_2|\psi_2\rangle\right) = c_1\hat{A}|\psi_1\rangle + c_2\hat{A}|\psi_2\rangle. \tag{3.3.2}$$

3.3.1 Properties of operators

1. **Equal operators:** Two operators $\hat{A}$ and $\hat{B}$ are equal if they produce the same state vector applied on arbitrary vector in the vector space $|\psi\rangle$,

$$\hat{A}|\psi\rangle = \hat{B}|\psi\rangle. \tag{3.3.3}$$

2. **Null operator:** Operator $\hat{A}$ is a null operator if it results in zero when acting on any state vector in the vector space,

$$\hat{A}|\psi\rangle = |0\rangle \equiv 0. \tag{3.3.4}$$

3. **Operator addition:** Given three operators $\hat{A}$, $\hat{B}$, and $\hat{C}$, addition of operators is associative,

$$\hat{A} + \left(\hat{B} + \hat{C}\right) = \left(\hat{A} + \hat{B}\right) + \hat{C} \tag{3.3.5}$$

and commutative,

$$\hat{A} + \hat{B} = \hat{B} + \hat{A}. \tag{3.3.6}$$

4. **Operator multiplication:** Given three operators $\hat{A}$, $\hat{B}$, and $\hat{C}$, multiplication of operators is associative,

$$\hat{A}\left(\hat{B}\hat{C}\right) = \left(\hat{A}\hat{B}\right)\hat{C}. \tag{3.3.7}$$

but not commutative, in general,

$$\hat{A}\hat{B} \neq \hat{B}\hat{A}. \tag{3.3.8}$$

3.3.2 Hermitian conjugate (or Hermitian adjoint) operator in the dual space

Given operator $\hat{A}$, which acts on vector $|\psi\rangle$ in the vector space from the left as

$$\hat{A}|\psi\rangle, \tag{3.3.9}$$

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the Hermitian adjoint operator $\hat{A}^\dagger$ acts on the dual vector $\langle\psi|$ from the right as

$$\langle\psi|\hat{A}^\dagger, \tag{3.3.10}$$

where $\hat{A}^\dagger$ is defined as the transpose and complex conjugate of $\hat{A}$,

$$\hat{A}^\dagger = (\hat{A})^{T*} \tag{3.3.11}$$

There is a distinction in the interpretations of $\hat{A}|\psi\rangle$ and $\langle\psi|\hat{A}^\dagger$. The operator $\hat{A}$ in $\hat{A}|\psi\rangle$ acts to the right on $|\psi\rangle$ while the operator $\hat{A}^\dagger$ acts to the left on $\langle\psi|$.

The Hermitian conjugate of the vector $\hat{A}|\psi\rangle$ itself is then

$$\left(\hat{A}|\psi\rangle\right)^\dagger = |\psi\rangle^\dagger \hat{A}^\dagger = \langle\psi|\hat{A}^\dagger, \tag{3.3.12}$$

where

$$\langle\psi| = |\psi\rangle^\dagger, \tag{3.3.13}$$

Taking the Hermitian conjugate again, $\langle\psi|^\dagger = (|\psi\rangle^\dagger)^\dagger = |\psi\rangle$; i.e.,

$$\langle\psi|^\dagger = |\psi\rangle. \tag{3.3.14}$$

Thus, a bra vector is the Hermitian conjugate of a ket vector, and vice versa.

Note that for a product of operators, $(\hat{A}\hat{B})^\dagger = (\hat{A}\hat{B})^{T*} = (\hat{B}^T\hat{A}^T)^* = \hat{B}^{T*}\hat{B}^{T*} = \hat{B}^\dagger\hat{A}^\dagger$. Furthermore given $c\hat{A}$, where c is a complex number, we have $(c\hat{A})^\dagger = c^*\hat{A}^\dagger$.

3.3.3 Outer product operators

Given two state vectors $|\psi\rangle$ and $|\phi\rangle$, we can construct an operator using the outer product

$$|\psi\rangle\langle\phi|. \tag{3.3.15}$$

This operator can then act on a state vector $|\chi\rangle$ as

$$[|\psi\rangle\langle\phi|]\,|\chi\rangle = |\psi\rangle\langle\phi|\chi\rangle. \tag{3.3.16}$$

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Taking the Hermitian conjugate,

$$(|\psi\rangle\langle\phi|\chi\rangle)^\dagger = \langle\chi|\phi\rangle\langle\psi|. \tag{3.3.17}$$

On the other hand, noting that $\langle\phi|\chi\rangle$ is a complex number, we rewrite the Hermitian conjugate of (3.3.16) as

$$(|\psi\rangle\langle\phi|\chi\rangle)^\dagger = \langle\phi|\chi\rangle^*\langle\psi|. \tag{3.3.18}$$

Equating (3.3.17) and (3.3.18),

$$\langle\chi|\phi\rangle = \langle\phi|\chi\rangle^*. \tag{3.3.19}$$

This also follows directly noting that $\langle\phi|\chi\rangle$ is a complex number and taking the Hermitian conjugate is equivalent to taking the complex conjugate, since transposing has no effect on a number.

3.3.4 Matrix elements of operators

The matrix element of operator $\hat{A}$ between states $|\phi\rangle$ and $|\psi\rangle$ is defined as

$$\langle\phi|\hat{A}|\psi\rangle. \tag{3.3.20}$$

Because the matrix element is a complex number, its Hermitian conjugate is equivalent to its complex conjugate, and taking the Hermitian conjugate, we have

$$\langle\phi|\hat{A}|\psi\rangle^\dagger = \langle\psi|\hat{A}^\dagger|\phi\rangle = \langle\phi|\hat{A}|\psi\rangle^*, \tag{3.3.21}$$

or

$$\langle\psi|\hat{A}^\dagger|\phi\rangle = \langle\phi|\hat{A}|\psi\rangle^*, \tag{3.3.22}$$

which relates the matrix elements of $\hat{A}^\dagger$ and $\hat{A}$.

3.3.5 Hermitian operators

A Hermitian operator $\hat{A}$ is one for which the Hermitian conjugate is identical to the operator,

$$\hat{A}^\dagger = \hat{A}. \tag{3.3.23}$$

In other words, given $\hat{A}|\psi\rangle$ in the vector space, we have $\langle\psi|\hat{A}^\dagger = \langle\psi|\hat{A}$ in the dual vector space.

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3.3.6 Matrix elements of a Hermitian operator

Question 3.3.2: For matrix elements of a Hermitian operator $\hat{A}$, (A). $\langle\phi|\hat{A}|\psi\rangle^* = \langle\psi|\hat{A}|\phi\rangle$, (B). $\langle\phi|\hat{A}|\psi\rangle^* = \langle\phi|\hat{A}|\psi\rangle$, (C). $\langle\phi|\hat{A}|\phi\rangle = \langle\psi|\hat{A}|\psi\rangle$, (D). $\langle\phi|\hat{A}|\psi\rangle = \langle\phi|\hat{A}^\dagger|\psi\rangle$.

Given a Hermitian operator $\hat{A}$, its matrix between states $|\phi\rangle$ and $|\psi\rangle$ satisfies, noting that $\hat{A}^\dagger = \hat{A}$ and that $\langle\phi|\hat{A}|\psi\rangle$ is a number , $\langle\phi|\hat{A}|\psi\rangle^\dagger = \langle\phi|\hat{A}|\psi\rangle^* = \langle\psi|\hat{A}^\dagger|\phi\rangle = \langle\psi|\hat{A}|\phi\rangle$; i.e.,

$$\langle\phi|\hat{A}|\psi\rangle^* = \langle\psi|\hat{A}|\phi\rangle. \tag{3.3.24}$$

In particular, taking $|\phi\rangle = |\psi\rangle$, the expectation value of $\hat{A}$ in state $|\psi\rangle$ satisfies

$$\langle\psi|\hat{A}|\psi\rangle^* = \langle\psi|\hat{A}|\psi\rangle \equiv \langle\hat{A}\rangle. \tag{3.3.25}$$

In other words, a Hermitian operator has real expectation values. Observe that $\hat{A}^\dagger = \hat{A}$ also means that

$$\langle\phi|\hat{A}|\psi\rangle = \langle\phi|\hat{A}^\dagger|\psi\rangle \tag{3.3.26}$$

In other words, we get the same result whether $\hat{A}$ acts to the right or to the left.

3.3.7 Eigenvalues and eigenvector of Hermitian operators

If an operator $\hat{A}$ acting on a state vectors $|a_n\rangle$ produces a number a_n times the same vector as

$$\hat{A}|a_n\rangle = a_n|a_n\rangle \tag{3.3.27}$$

we say $|a_n\rangle$ is an eigenvector of $\hat{A}$ with eigenvalue a_n. For a Hermitian operator $\hat{A}$, the left hand side of (3.3.24) with $|\psi\rangle = |a_n\rangle$ and $|\phi\rangle = |a_m\rangle$ gives $\langle a_m|\hat{A}|a_n\rangle^* = a_n^*\langle a_m|a_n\rangle^* = a_n^*\langle a_n|a_m\rangle$, and the right hand gives $\langle a_n|\hat{A}|a_m\rangle = a_m\langle a_n|a_m\rangle$. Equating these,

$$(a_n^* - a_m)\langle a_n|a_m\rangle = 0. \tag{3.3.28}$$

Therefore, for $m = n$, we have $\langle a_n|a_n\rangle \neq 0$ and $a_n^* = a_n$. In other words, the eigenvalues are real.

For $m \neq n$, and assuming nondegenerate states, such that different eigenstates correspond to different eigenvalues, we have $a_n^* - a_m \neq 0$ and $\langle a_m|a_n\rangle = 0$. In other words, the eigenvectors are

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orthogonal. With normalized eigenvectors, we then have

$$\langle a_m|a_n\rangle = \delta_{mn}. \tag{3.3.29}$$

If the system has degenerate eigenvectors, different eigenvectors corresponding to the same eigenvalues, one uses the Gram-Schmidt orthogonalization to construct an orthonormal set of eigenvectors.

3.3.8 Completeness of eigenvectors of a Hermitian operator

Given the eigenvectors $|a_n\rangle$ of a Hermitian operator $\hat{A}$ for a system, can we express an arbitrary normalized state $|\psi\rangle$ of the system as a linear superposition of $|a_n\rangle$ as $|\psi\rangle = \sum_n c_n|a_n\rangle$? Is the set of $|a_n\rangle$ complete? Consider a normalized state $|\psi\rangle$ and define

$$|\Psi\rangle \equiv |\psi\rangle - \sum_n |a_n\rangle\langle a_n|\psi\rangle. \tag{3.3.30}$$

Because $|\psi\rangle$ is normalized,

$$\begin{aligned}
\langle\Psi|\Psi\rangle &= \left(\langle\psi| - \sum_m \langle\psi|a_m\rangle\langle a_m|\right)\left(|\psi\rangle - \sum_n |a_n\rangle\langle a_n|\psi\rangle\right) \\
&= \langle\psi|\psi\rangle - \sum_m \langle\psi|a_m\rangle\langle a_m|\psi\rangle - \sum_n \langle\psi|a_n\rangle\langle a_n|\psi\rangle + \sum_{nm}\langle\psi|a_m\rangle\delta_{mn}\langle a_n|\psi\rangle \\
&= \langle\psi|\psi\rangle - \sum_n \langle\psi|a_n\rangle\langle a_n|\psi\rangle \qquad (3.3.31) \\
&= \langle\psi|\psi\rangle - \sum_n |\langle a_n|\psi\rangle|^2 \qquad (3.3.32) \\
&= 1 - 1 = 0, \qquad (3.3.33)
\end{aligned}$$

since $\langle\psi|\psi\rangle = 1$ for a normalized $|\psi\rangle$ and $\sum_n |\langle a_n|\psi\rangle|^2 = 1$ for $|a_n\rangle$ containing all outcomes and summing over all n gives a total probability of one. Therefore, the set of $|a_n\rangle$ is complete if the sum of $|\langle a_n|\psi\rangle|^2$ equals one. In other words, if the state vector of every possible measurement outcome of $\hat{A}$ is included, since the total probability must be one. Looking at the next to last line in the equation above, it also means that

$$\sum_n |a_n\rangle\langle a_n| = 1; \tag{3.3.34}$$

i.e., $\sum_n |a_n\rangle\langle a_n|$ is an identity operator.

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To clarify the notation, let us take the familiar example of a vector in 3-dimensions, $\vec{r} = (x, y, z)$. In our bra-ket notation, $|r\rangle = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = x|1\rangle + x|2\rangle + x|3\rangle$, where $|1\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $|2\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$, and $|3\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$. Therefore, $x = \langle 1|r\rangle$, $y = \langle 2|r\rangle$, and $z = \langle 3|r\rangle$, or $|r\rangle = |1\rangle\langle 1|r\rangle + |2\rangle\langle 2|r\rangle + |3\rangle\langle 3|r\rangle$, where $|1\rangle\langle 1|r\rangle$ projects out the $x-$component together with its direction $|r\rangle$. In particular, $|1\rangle\langle 1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}\begin{pmatrix} 1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$. Therefore, $|1\rangle\langle 1| + |2\rangle\langle 2| + |3\rangle\langle 3| = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = 1$, which is the completeness relation, and $|r\rangle = |1\rangle\langle 1|r\rangle + |2\rangle\langle 2|r\rangle + |3\rangle\langle 3|r\rangle = (|1\rangle\langle 1| + |2\rangle\langle 2| + |3\rangle\langle 3|)\,|r\rangle = |r\rangle$. Observe that the completeness $\sum_n |n\rangle\langle n| = 1$ allows us to decompose and write any vector in 3-dimension in terms the basis vectors $|n\rangle$, $n = 1, 2, 3$.

3.3.9 Continuous spectra of a Hermitian operator

Above, we have been looking only at discrete spectra in which the eigenvalues of $\hat{A}$ take discrete values. Now consider continuous eigenvalues of a Hermitian operator $\hat{A}$ such that

$$\hat{A}|a\rangle = a|a\rangle, \tag{3.3.35}$$

where a takes continuous values, such as the position of a particle or the momentum of a free particle. Taking the matrix element of $\hat{A}$ between states $|a'\rangle$ and $|a\rangle$, we have

$$\langle a'|\hat{A}|a\rangle = a\langle a'|a\rangle = a'\langle a'|a\rangle, \tag{3.3.36}$$

where we have applied $\hat{A}$ on the right for the first equality and on the left for the last equality, since it is Hermitian. Integrating both sides over all values of a', which takes continuum values, we have

$$\int da'\, \langle a'|\hat{A}|a\rangle = a \int da'\, \langle a'|a\rangle = \int da'\, a'\langle a'|a\rangle, \tag{3.3.37}$$

which holds if

$$\langle a'|a\rangle = \delta(a' - a). \tag{3.3.38}$$

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This is the orthonormality condition of the state vectors for continuous spectra of a Hermitian operator. They are normalized through the delta function. The completeness condition, which was $\sum_n |a_n\rangle\langle a_n| = 1$ for discrete spectra, now becomes

$$\int da\, |a\rangle\langle a| = 1. \tag{3.3.39}$$

Alternatively, we can start with the completeness relation, $\hat{I} = \sum_n |a_n\rangle\langle a_n| = 1$ for discrete eigenvalues, which becomes $\hat{I} = \int da\, |a\rangle\langle a| = 1$. First, for the discrete case, squaring $1 = \hat{I}^2 = \sum_{n,m} |a_n\rangle\langle a_n|a_m\rangle\langle a_m| = \sum_{n,m} \delta_{nm}|a_n\rangle\langle a_m| = \sum_n |a_n\rangle\langle a_n|$. It is the case for a projection operator that $\hat{I}^2 = \hat{I}$. Requiring the same for the case of continuous eigenvalues,

$$\hat{I}^2 = \int da'\, da\, |a'\rangle\langle a'|a\rangle\langle a| = \int da\, |a\rangle\langle a| = \hat{I}, \tag{3.3.40}$$

leads to $\langle a'|a\rangle = \delta(a' - a)$.

3.3.10 Gram-Schmidt orthogonalization procedure

When two or more states correspond to the same eigenvalue, we say the states are degenerate. If there are n states with the same eigenvalue, we say the eigenvalue, such as an energy level, is $n-$fold degenerate (or there are $n-$fold degenerate states).

Consider an $n-$fold degenerate eigenvalue a for an operator $\hat{A}$, with states $|a_i\rangle$, $i = 1, 2, ..., n$ such that $\hat{A}|a_i\rangle = a|a_i\rangle$. Based on our investigation, the eigenstates would be orthogonal when there is no degeneracy. Now that we have degeneracy, how do we ensure that the states are orthogonal? The Gram-Schmidt procedure is a recipe for constructing the orthogonal states.

1. Normalize each state and define $\alpha_i = \frac{|a_i\rangle}{\sqrt{\langle a_i|a_i\rangle}}$.
2. Start with one of the normalized states, $|\alpha_1'\rangle = |\alpha_1\rangle$, as one of the orthonormal states.
3. Subtract the component of $|\alpha_2\rangle$ along $|\alpha_1\rangle$, $|\alpha_2\rangle - |\alpha_1\rangle\langle\alpha_1|\alpha_2\rangle$, normalize and promote it to $|\alpha_2'\rangle = \frac{|\alpha_2\rangle - |\alpha_1\rangle\langle\alpha_1|\alpha_2\rangle}{\sqrt{\langle\alpha_2|\alpha_2\rangle - \langle\alpha_1|\alpha_2\rangle\langle\alpha_2|\alpha_1\rangle}} = \frac{|\alpha_2\rangle - |\alpha_1\rangle\langle\alpha_1|\alpha_2\rangle}{\sqrt{1 - \langle\alpha_1|\alpha_2\rangle\langle\alpha_2|\alpha_1\rangle}}$. Observe that $\langle\alpha_2'|\alpha_2'\rangle = 1$ and $\langle\alpha_1'|\alpha_2'\rangle = 0$.
4. Subtract the components of $|\alpha_3\rangle$ along $|\alpha_1\rangle$ and $|\alpha_2\rangle$, $|\alpha_3\rangle - |\alpha_1\rangle\langle\alpha_1|\alpha_2\rangle - |\alpha_2\rangle\langle\alpha_2|\alpha_2\rangle$, normalize and promote it to $|\alpha_3'\rangle$.
5. Continue until you exhaust all n states.

Exercise 3.3.1: Check that the states that result from the Gram-Schmidt or-

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thogonalization procedure are orthonormal.

3.4 Position and momentum representations

Consider the position operator $\hat{\vec{r}}$. Let us denote the eigenstate corresponding to eigenvalue (or position $\vec{r}$) by $|\vec{r}\rangle$; i.e.,

$$\hat{\vec{r}}|\vec{r}\rangle = \vec{r}|\vec{r}\rangle. \tag{3.4.1}$$

Because $\vec{r}$ takes continuous values, these states are normalized through the delta function as

$$\langle\vec{r}'|\vec{r}\rangle = \delta^3(\vec{r}' - \vec{r}) \tag{3.4.2}$$

and the identity operator is

$$\int d^3\vec{r}\,|\vec{r}\rangle\langle\vec{r}| = 1. \tag{3.4.3}$$

Observe that we have a continuum infinite dimensional vector space. Note that for the matrix element of $\hat{\vec{r}}$,

$$\langle\vec{r}'|\hat{\vec{r}}|\vec{r}\rangle = \vec{r}\delta^3(\vec{r}' - \vec{r}). \tag{3.4.4}$$

Similarly, for a free particle in one dimension, which can have continuous values of momentum, denoting the eigenstate corresponding to momentum $\vec{p}$ by $|\vec{p}\rangle$,

$$\hat{\vec{p}}|\vec{p}\rangle = \vec{p}|\vec{p}\rangle, \tag{3.4.5}$$

$$\langle\vec{p}'|\vec{p}\rangle = \delta^3(\vec{p}' - \vec{p}) \tag{3.4.6}$$

and the identity operator is

$$\int d^3\vec{p}\,|\vec{p}\rangle\langle\vec{p}| = 1. \tag{3.4.7}$$

The expectation value of $\hat{\vec{r}}$ in state $|\psi(t)\rangle$ in position space is then

$$\begin{aligned}
\langle\psi(t)|\hat{\vec{r}}|\psi(t)\rangle &= \langle\psi(t)|\hat{\vec{r}}\left(\int d^3\vec{r}|\vec{r}\rangle\langle\vec{r}|\right)|\psi(t)\rangle = \int d^3\vec{r}\langle\psi(t)|\hat{\vec{r}}|\vec{r}\rangle\langle\vec{r}|\psi(t)\rangle \\
&= \int d^3\vec{r}\,\vec{r}\langle\psi(t)|\vec{r}\rangle\langle\vec{r}|\psi(t)\rangle = \int d^3\vec{r}\,\vec{r}\langle\langle\vec{r}|\psi(t)\rangle\psi(\vec{r},t) \\
&= \int d\vec{r}\,\vec{r}|\psi(\vec{r},t)|^2 = \int d^3\vec{r}\,\vec{r}\mathcal{P}(\vec{r},t),
\end{aligned} \tag{3.4.8}$$

where $\psi(\vec{r},t) = \langle\vec{r}|\psi(t)\rangle$ and $\psi(\vec{r},t)^* = \langle\psi(t)|\vec{r}\rangle$. Therefore, given a state vector $|\psi(t)\rangle$, the corre-

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sponding wavefunction in position space as

$$\langle \vec{r}|\psi(t)\rangle \equiv \psi(\vec{r},t). \tag{3.4.9}$$

In the equation $\hat{\vec{r}}|\vec{r}\rangle = \vec{r}|\vec{r}\rangle$, $\hat{\vec{r}}$ is an operator, $|\vec{r}\rangle$ is a state vector, and $\vec{r}$ is an eigenvalue. In position representation, the eigenvalue $\vec{r}$ has real components by definition, since it represents the coordinates of a point in real 3-dimensional space. In other words, $\vec{r}^* = \vec{r}$.

Question 3.4.1: $\psi(\vec{r},t)$ is (A). a vector, (B). a scalar.

With the expectation values of $\hat{\vec{r}}$ and $\hat{\vec{p}}$ in state $|\psi\rangle$ related by

$$\langle\psi|\hat{\vec{p}}|\psi\rangle = \frac{d}{dt}\langle\psi|\hat{\vec{r}}|\psi\rangle. \tag{3.4.10}$$

the representation of the momentum operator in position space then follows from (1.11.3),

$$\hat{\vec{p}} = -i\hbar\vec{\nabla}. \tag{3.4.11}$$

To find the matrix element $\langle x|\hat{p}|x'\rangle$, consider the expectation value of the momentum operator in position representation is

$$\begin{aligned}
\langle\psi|\hat{p}|\psi\rangle &= m\frac{d}{dt}\langle\psi|\hat{x}|\psi\rangle = \int dx\langle\psi(t)|x\rangle(-i\hbar\frac{\partial}{\partial x})\langle x|\psi(t)\rangle \\
&= \int dx\int dx'\langle\psi(t)|x\rangle(-i\hbar\frac{\partial}{\partial x})\langle x|x'\rangle\langle x'|\psi(t)\rangle \\
&= \int dx\int dx'\langle\psi(t)|x\rangle(-i\hbar\frac{\partial}{\partial x})\delta(x-x')\langle x'|\psi(t)\rangle \\
&= \int dx\int dx'\langle\psi(t)|x\rangle\langle x|\hat{p}|x'\rangle\langle x'|\psi(t)\rangle;
\end{aligned} \tag{3.4.12}$$

i.e.,

$$\langle x|\hat{p}|x'\rangle = -i\hbar\frac{\partial}{\partial x}\delta(x-x') = -i\hbar\frac{\partial}{\partial x}\langle x|x'\rangle. \tag{3.4.13}$$

Alternatively,

$$\begin{aligned}
\langle\psi|\hat{p}|\psi\rangle &= \int dxdx'\langle\psi|x\rangle\langle x|\hat{p}|x'\rangle\langle x'|\psi\rangle \\
&= \int dxdx'dp\langle\psi|x\rangle\langle x|\hat{p}|p\rangle\langle p|x'\rangle\langle x'|\psi\rangle
\end{aligned}$$

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$$
\begin{aligned}
&= \int dxdx'\,dp\; p\langle\psi|x\rangle\langle x|p\rangle\langle p|x'\rangle\langle x'|\psi\rangle \\
&= \int dxdx'\,dp\; p\langle\psi|x\rangle\frac{1}{\sqrt{2\pi\hbar}}e^{\frac{i}{\hbar}px}\frac{1}{\sqrt{2\pi\hbar}}e^{-\frac{i}{\hbar}px'}\langle x'|\psi\rangle \\
&= \int dxdx'\langle\psi|x\rangle\left(\frac{1}{2\pi\hbar}\int dp\, p e^{\frac{i}{\hbar}p(x-x')}\right)\langle x'|\psi\rangle \\
&= \int dxdx'\langle\psi|x\rangle\left(-i\hbar\frac{\partial}{\partial x}\delta(x-x')\right)\langle x'|\psi\rangle. \qquad (3.4.14)
\end{aligned}
$$

Comparing the first and the last lines, we have again, $\langle x|\hat{p}|x'\rangle = -i\hbar\frac{\partial}{\partial x}\delta(x-x')$, and the position operator in momentum representation is $\hat{p} = -i\hbar\frac{\partial}{\partial x}$.

Exercise 3.4.1: Show that $\hat{p} = -i\hbar\frac{\partial}{\partial x}$ is Hermitian.

Solution:

Method1: Using integration by parts.

$$
\begin{aligned}
\langle\phi|\hat{p}|\psi\rangle &= \int dx\langle\phi|x\rangle\langle x|\hat{p}|\psi\rangle = \int dx\phi(x)^*\left(-i\hbar\frac{\partial}{\partial x}\psi(x)\right) \\
&= -i\hbar\int dx\left[\frac{\partial}{\partial x}\left(\phi(x)^*\psi(x)\right) - \frac{\partial}{\partial x}\phi(x)^*\psi(x)\right] \\
&= \int dx\left[i\hbar\frac{\partial}{\partial x}\phi(x)^*\psi(x)\right] = \int dx\left[\left(-i\hbar\frac{\partial}{\partial x}\phi(x)\right)^*\psi(x)\right] \\
&= \int dx\langle x|\hat{p}|\phi\rangle^*\langle x|\psi\rangle = \int dx\langle\phi|\hat{p}^\dagger|x\rangle\langle x|\psi\rangle = \langle\phi|\hat{p}^\dagger|\psi\rangle. \qquad (3.4.15)
\end{aligned}
$$

Therefore, $\hat{p}^\dagger = \hat{p}$, and $\hat{\vec{p}}^\dagger = \hat{\vec{p}}$ in three dimensions.

Method 2: Using the delta function. A bit more formally and not using $\langle x|\hat{p}|\psi\rangle = i\hbar\frac{\partial}{\partial x}\psi(x)$, even though that should be obvious at this point. It used a property of the delta function $\int dx\delta'(x-a)f(x) = f'(x)|_{x=a}$,

$$
\begin{aligned}
\langle\phi|\hat{p}|\psi\rangle &= \int dxdx'\langle\phi|x'\rangle\langle x'|\hat{p}|x\rangle\langle x|\psi\rangle = \int dxdx'\,\phi(x')^*\left(-i\hbar\frac{\partial}{\partial x'}\delta(x'-x)\right)\psi(x) \\
&= \int dx'\left(i\hbar\frac{\partial}{\partial x'}\phi(x')^*\right)\psi(x) = \int dx\left[\left(-i\hbar\frac{\partial}{\partial x}\phi(x)\right)^*\psi(x)\right] \\
&= \int dx\langle x|\hat{p}|\phi\rangle^*\langle x|\psi\rangle = \int dx\langle\phi|\hat{p}^\dagger|x\rangle\langle x|\psi\rangle = \langle\phi|\hat{p}^\dagger|\psi\rangle, \qquad (3.4.16)
\end{aligned}
$$

where we have used $\int\delta'(x)f(x)dx = -f'(x)$, which follows from integration by parts,

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$\int \delta'(x-x')f(x)dx = \int \left(\frac{d}{dx}(\delta(x-x')f(x)) - \delta(x-x')f'(x)\right)dx = -f'(x').$

Question 3.4.2: Is $\frac{\partial}{\partial x}$ is Hermitian? (A). Yes, (B). No.

To find the matrix element $\langle x|\hat{p}|p\rangle$,

$$\begin{aligned}\langle x|\hat{p}|p\rangle &= \int dx'\langle x|\hat{p}|x'\rangle\langle x'|p\rangle = \int dx'\left(-i\hbar\frac{\partial}{\partial x}\delta(x-x')\right)\langle x'|p\rangle = \int dx'\left(i\hbar\frac{\partial}{\partial x'}\delta(x-x')\right)\langle x'|p\rangle \\ &= -i\hbar\frac{\partial}{\partial x}\langle x|p\rangle; \end{aligned} \tag{3.4.17}$$

where we have used

$$\int \delta'(x)f(x)dx = -f'(x), \tag{3.4.18}$$

which follows from integration by parts, $\int \delta'(x-x')f(x)dx = \int \left(\frac{d}{dx}(\delta(x-x')f(x)) - \delta(x-x')f'(x)\right)dx = -f'(x')$. On the other hand, $\hat{p}|p\rangle = p|p\rangle$ gives $\langle x|\hat{p}|p\rangle = p\langle x|p\rangle$. Therefore, $\langle x|p\rangle$ is the wavefunction of momentum p and that $\langle x|\hat{p}|p\rangle = -i\hbar\frac{\partial}{\partial x}\langle x|p\rangle$ in position space, and

$$-i\hbar\frac{\partial}{\partial x}\langle x|p\rangle = p\langle x|p\rangle. \tag{3.4.19}$$

The solution is

$$\langle x|p\rangle = Ae^{\frac{i}{\hbar}px}, \tag{3.4.20}$$

where A a normalization constant. To find the normalization constant, recalling that $\langle p'|p\rangle = \delta(p'-p)$,

$$\begin{aligned}\delta(p'-p) &= \langle p'|p\rangle = \int dx\langle p'|x\rangle\langle x|p\rangle = \int dx(Ae^{\frac{i}{\hbar}p'x})^*Ae^{\frac{i}{\hbar}px} \\ &= |A|^2\int dxe^{-\frac{i}{\hbar}(p'-p)x} = |A^2|2\pi\delta(\frac{p'-p}{\hbar}) = |A|^2 2\pi\hbar\delta(p'-p) \end{aligned} \tag{3.4.21}$$

Therefore, $A = \frac{1}{\sqrt{2\pi\hbar}}$, and the normalized wavefunction of definite momentum p in position space is

$$\langle x|p\rangle = \frac{1}{\sqrt{2\pi\hbar}}e^{\frac{i}{\hbar}px}. \tag{3.4.22}$$

The three-dimensional expressions follow immediately. The identities become

$$\langle \vec{r}|\vec{p}\rangle = \frac{1}{(2\pi\hbar)^{3/2}}e^{\frac{i}{\hbar}\vec{p}\cdot\vec{r}}. \tag{3.4.23}$$

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With (3.4.22) in hand, we can write the normalized wavefunction of definite position in momentum space, $\langle p|x\rangle = \langle x|p\rangle^*$,

$$\langle p|x\rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{-\frac{i}{\hbar}px}. \tag{3.4.24}$$

and

$$\langle \vec{p}|\vec{r}\rangle = \frac{1}{(2\pi\hbar)^{3/2}} e^{-\frac{i}{\hbar}\vec{p}\cdot\vec{r}}. \tag{3.4.25}$$

Exercise 3.4.2: Show that the position operator in position representation is Hermitian.

Answer:

$$\begin{aligned}
\langle\phi|\hat{\vec{r}}^\dagger|\psi\rangle &= \langle\psi|\hat{\vec{r}}|\phi\rangle^\dagger = \langle\psi|\hat{\vec{r}}|\phi\rangle^* = \left(\langle\psi \int d^3\vec{r}\,|\vec{r}\rangle\langle\vec{r}|\hat{\vec{r}} \int d^3\vec{r}'\,|\vec{r}'\rangle\langle\vec{r}'|\phi\rangle\right)^* \\
&= \left(\int d^3\vec{r} \int d^3\vec{r}'\,\langle\psi\,|\vec{r}\rangle\langle\vec{r}|\hat{\vec{r}}|\vec{r}'\rangle\langle\vec{r}'|\phi\rangle\right)^* \\
&= \int d^3\vec{r} \int d^3\vec{r}'\,\langle\psi\,|\vec{r}\rangle^*\vec{r}'\,\delta^3(\vec{r}-\vec{r}')\,\langle\vec{r}'|\phi\rangle^* \\
&= \int d^3\vec{r}\langle\psi\,|\vec{r}\rangle^*\vec{r}\,\langle\vec{r}|\phi\rangle^* = \int d^3\vec{r}\langle\phi\,|\vec{r}\rangle\vec{r}\,\langle\vec{r}|\psi\rangle = \int d^3\vec{r}\langle\phi|\vec{r}|\vec{r}\rangle\,\langle\vec{r}|\psi\rangle \\
&= \langle\phi|\hat{\vec{r}}|\psi\rangle,
\end{aligned} \tag{3.4.26}$$

where we have set $\vec{r}^* = \vec{r}$ for the eigenvalue, since the eigenvalue is the position $\vec{r}$ which is real by definition, and the order of where we put the scalars $\langle\phi\,|\vec{r}\rangle$ and $\langle\vec{r}|\psi\rangle$ can be chosen at will. Therefore, $\langle\phi|\hat{\vec{r}}^\dagger|\psi\rangle = \langle\phi|\hat{\vec{r}}|\psi\rangle$ and $\hat{\vec{r}}$ is Hermitian.

3.4.1 Operators and wavefunction in position representation

Now suppose the state vector of a system in Hilbert space is given by $|\psi(t)\rangle$. The position and the momentum operators in position space are given by

$$\hat{\vec{r}} = \vec{r}, \qquad \hat{\vec{p}} = -i\hbar\vec{\nabla}. \tag{3.4.27}$$

A physical observable in position representation is a function of position and momentum operators in position representation,

$$\hat{O}(\hat{\vec{r}}, \hat{\vec{p}}) = \hat{O}(\vec{r}, -i\hbar\vec{\nabla}). \tag{3.4.28}$$

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The expectation value $\hat{O}$ is a measurable quantity that should be independent of the representation. In state $\psi(t)$,

$$\langle \hat{O} \rangle = \langle \psi(t) | \hat{O} | \psi(t) \rangle. \tag{3.4.29}$$

To write the expectation value in position representation, we insert $\int d^2\vec{r}\, |\vec{r}\rangle\langle\vec{r}| = 1$, and we have

$$\langle \hat{O} \rangle = \int d^2\vec{r} \langle \psi(t) | \vec{r} \rangle \langle \vec{r} | \hat{O} | \psi(t) \rangle. \tag{3.4.30}$$

Thus,

$$\langle \psi(t) | \vec{r} \rangle = \langle \vec{r} | \psi(t) \rangle^* = \psi(\vec{r}, t)^* \tag{3.4.31}$$

and $\langle \vec{r} | \hat{O} | \psi(t) \rangle$ is just $\hat{O} | \psi(t) \rangle$ in position representation; i.e.,

$$\langle \vec{r} | \hat{O} | \psi(t) \rangle = \hat{O}(\vec{r}, -i\hbar\vec{\nabla}) \psi(\vec{r}, t) \tag{3.4.32}$$

so that

$$\langle \hat{O} \rangle = \int d^3\vec{r} \psi(\vec{r}, t)^* \hat{O}(\vec{r}, -i\hbar\vec{\nabla}) \psi(\vec{r}, t). \tag{3.4.33}$$

In other words, given a state vector $|\psi(t)\rangle$ in Hilbert space, the wavefunction in position representation is

$$\psi(\vec{r}, t) = \langle \vec{r} | \psi(t) \rangle. \tag{3.4.34}$$

Question 3.4.3: Considering the unit vectors in three dimensions, what does $\hat{x}(\hat{x}\cdot) + \hat{y}(\hat{y}\cdot) + \hat{z}(\hat{z}\cdot)$ represent?

Question 3.4.4: Write down and discuss the wavefunctions of definite position and definite momentum in both position and momentum representation.

Question 3.4.5: Why does he negative sign in the exponent in $\langle p | x \rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{-\frac{i}{\hbar}px}$ indicate in contrast to the positive sign in contrast to $\langle x | p \rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{\frac{i}{\hbar}px}$.

3.4.2 Operators and wavefunction in momentum representation

First, the momentum operator in momentum space is $\hat{\vec{p}} = \vec{p}$. To determine the position operator in momentum space, we seek an operator $\hat{r}$ which produces a definite position when applied on

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a wavefunction of definite position in momentum space; i.e.,

$$\hat{r}\frac{1}{(2\pi\hbar)^{3/2}}e^{-\frac{i}{\hbar}\vec{p}\cdot\vec{r}} = \vec{r}\frac{1}{(2\pi\hbar)^{3/2}}e^{-\frac{i}{\hbar}\vec{p}\cdot\vec{r}}, \tag{3.4.35}$$

and, therefore, the position operator in momentum space is given by

$$\hat{\vec{r}} = i\hbar\vec{\nabla}_{\vec{p}}, \tag{3.4.36}$$

where $\vec{\nabla}_{\vec{p}}$ is the gradient operator with respect to momentum. In particular, in one dimension,

$$\hat{x} = i\hbar\frac{\partial}{\partial p}, \tag{3.4.37}$$

In $\hat{\vec{p}}|\vec{p}\rangle = \vec{p}|\vec{p}\rangle$, $\hat{\vec{p}}$ is an operator, $|\vec{p}\rangle$ is a state vector, and $\vec{p}$ is an eigenvalue. In momentum representation, the eigenvalue $\vec{p}$ has real components by definition, since it represents the momentum of a particle. In other words, $\vec{p}^{\,*} = \vec{p}$.

Exercise 3.4.3: Show that the position operator in momentum representation is given by $\hat{x} = i\hbar\frac{\partial}{\partial p}$ directly using matrix elements.

Solution:

$$\begin{aligned}
\langle\psi|\hat{x}|\psi\rangle &= \int dpdp'\langle\psi|p\rangle\langle p|\hat{x}|p'\rangle\langle p'|\psi\rangle \\
&= \int dpdp'dx\langle\psi|p\rangle\langle p|x\rangle\langle x|\hat{x}|p'\rangle\langle p'|\psi\rangle \\
&= \int dpdp'dx\, x\langle\psi|p\rangle\langle p|x\rangle\langle x|p'\rangle\langle p'|\psi\rangle \\
&= \int dpdp'dx\, x\langle\psi|p\rangle\frac{1}{\sqrt{2\pi\hbar}}e^{-\frac{i}{\hbar}px}\frac{1}{\sqrt{2\pi\hbar}}e^{\frac{i}{\hbar}p'x}\langle p'|\psi\rangle \\
&= \int dpdp'\langle\psi|p\rangle\left(\frac{1}{2\pi\hbar}\int dx\, xe^{-\frac{i}{\hbar}(p-p')x}\right)\langle p'|\psi\rangle \\
&= \int dpdp'\langle\psi|p\rangle\left(i\hbar\frac{\partial}{\partial p}\delta(p-p')\right)\langle p'|\psi\rangle.
\end{aligned} \tag{3.4.38}$$

Comparing the first and the last lines, we have

$$\langle p|\hat{x}|p'\rangle = i\hbar\frac{\partial}{\partial p}\delta(p-p'). \tag{3.4.39}$$

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Thus, the position operator in momentum representation is given by $\hat{x} = i\hbar\frac{\partial}{\partial p}$.

Exercise 3.4.4: Show that the position operator in momentum space is Hermitian.

Solution: The solution is simply a repeat of Exercise 3.4.1 with $p \leftrightarrow x$ and $i \leftrightarrow -i$.

An observable in momentum representation is a function of position and momentum operators in momentum representation,

$$\hat{O}(\hat{\vec{r}}, \hat{\vec{p}}) = \hat{O}(i\hbar\vec{\nabla}_{\vec{p}}, \vec{p}). \tag{3.4.40}$$

Because the expectation value $\hat{O}$ is independent of the representation, in state $|\psi(t)\rangle$,

$$\langle\hat{O}\rangle = \langle\psi(t)|\hat{O}|\psi(t)\rangle. \tag{3.4.41}$$

Inserting $\int d^2\vec{p}\,|\vec{p}\rangle\langle\vec{p}| = 1$, and we have

$$\langle\hat{O}\rangle = \int d^2\vec{p}\langle\psi(t)|\vec{p}\rangle\langle\vec{p}|\hat{O}|\psi(t)\rangle. \tag{3.4.42}$$

Thus,

$$\langle\psi(t)|\vec{p}\rangle = \langle\vec{p}|\psi(t)\rangle^* = \psi(\vec{p}, t)^* \tag{3.4.43}$$

and $\langle\vec{p}|\hat{O}|\psi(t)\rangle$ is just $\hat{O}|\psi(t)\rangle$ in momentum representation; i.e.,

$$\langle\vec{r}|\hat{O}|\psi(t)\rangle = \hat{O}(i\hbar\vec{\nabla}_{\vec{p}}, \vec{p}) \tag{3.4.44}$$

so that

$$\langle\hat{O}\rangle = \int d^3\vec{p}\psi(\vec{p}, t)^*\hat{O}(i\hbar\vec{\nabla}_{\vec{p}}, \vec{p})\psi(\vec{p}, t). \tag{3.4.45}$$

In other words, given a state vector $|\psi(t)\rangle$ in Hilbert space, the wavefunction in momentum representation is

$$\psi(\vec{p}, t) = \langle\vec{p}|\psi(t)\rangle. \tag{3.4.46}$$

Because both $\hat{\vec{r}}$ and $\hat{\vec{p}}$ are Hermitian and time is not an operator, a physical observable $\hat{O}(\hat{\vec{r}}, \hat{\vec{p}}, t)$ that is a function of $\hat{\vec{r}}$ and $\hat{\vec{p}}$ is Hermitian too. In particular, the Hamiltonian operator for energy $\hat{O}(\hat{\vec{r}}, \hat{\vec{p}}, t)$ is Hermitian.

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Exercise 3.4.5: Show that the Schrödinger $i\hbar\frac{\partial}{\partial t}|\psi(t)\rangle = \hat{H}|\psi(t)\rangle$, where $\hat{H} = \frac{\hat{\vec{p}}^2}{2m} + V(\hat{\vec{r}}, t)$ becomes

1. $$i\hbar\frac{\partial\psi(\vec{r},t)}{\partial t} = -\frac{\hbar^2}{2m}\vec{\nabla}^2\psi(\vec{r},t) + V(\vec{r},t)\psi(\vec{r},t) \tag{3.4.47}$$
   in position representation, and

2. $$i\hbar\frac{\partial\psi(\vec{p},t)}{\partial t} = -\frac{\vec{p}^2}{2m}\psi(\vec{p},t) + V(i\hbar\vec{\nabla}_{\vec{p}},t)\psi(\vec{p},t) \tag{3.4.48}$$
   in momentum representation.

Solution:

1. Inserting $1 = \int d^2\vec{r}|\vec{r}\rangle\langle\vec{r}|$ in $i\hbar\frac{\partial}{\partial t}|\psi(t)\rangle = \hat{H}|\psi(t)\rangle$ we have $i\hbar\frac{\partial}{\partial t}\int d^2\vec{r}|\vec{r}\rangle\langle\vec{r}|\psi(t)\rangle = \int d^2\vec{r}|\vec{r}\rangle\langle\vec{r}|\frac{\hat{\vec{p}}^2}{2m}|\psi(t)\rangle + \int d^2\vec{r}|\vec{r}\rangle\langle\vec{r}|V(\hat{\vec{r}},t)|\psi(t)\rangle$. Therefore, $\int d^2\vec{r}|\vec{r}\rangle\Big(i\hbar\frac{\partial}{\partial t}\langle\vec{r}|\psi(t)\rangle - \langle\vec{r}|\frac{\hat{\vec{p}}^2}{2m}|\psi(t)\rangle - \langle\vec{r}|V(\hat{\vec{r}},t)|\psi(t)\rangle\Big) = 0$. Therefore,

$$i\hbar\frac{\partial}{\partial t}\langle\vec{r}|\psi(t)\rangle = \langle\vec{r}|\frac{\hat{\vec{p}}^2}{2m}|\psi(t)\rangle + \langle\vec{r}|V(\hat{\vec{r}},t)|\psi(t)\rangle, \tag{3.4.49}$$

   or writing the position representations explicitly,

$$i\hbar\frac{\partial}{\partial t}\psi(\vec{r},t) = -\frac{\hbar^2\vec{\nabla}^2}{2m}\psi(\vec{r},t) + V(\vec{r},t)\psi(\vec{r},t). \tag{3.4.50}$$

2. Inserting $1 = \int d^2\vec{p}|\vec{p}\rangle\langle\vec{p}|$ in $i\hbar\frac{\partial}{\partial t}|\psi(t)\rangle = \hat{H}|\psi(t)\rangle$ we have $i\hbar\frac{\partial}{\partial t}\int d^2\vec{p}|\vec{p}\rangle\langle\vec{p}|\psi(t)\rangle = \int d^2\vec{p}|\vec{p}\rangle\langle\vec{r}|\frac{\hat{\vec{p}}^2}{2m}|\psi(t)\rangle + \int d^2\vec{p}|\vec{p}\rangle\langle\vec{p}|V(\hat{\vec{p}},t)|\psi(t)\rangle$. Therefore, $\int d^2\vec{p}|\vec{p}\rangle\Big(i\hbar\frac{\partial}{\partial t}\langle\vec{p}|\psi(t)\rangle - \langle\vec{p}|\frac{\hat{\vec{p}}^2}{2m}|\psi(t)\rangle - \langle\vec{p}|V(\hat{\vec{r}},t)|\psi(t)\rangle\Big) = 0$. Therefore,

$$i\hbar\frac{\partial}{\partial t}\langle\vec{p}|\psi(t)\rangle = \langle\vec{p}|\frac{\hat{\vec{p}}^2}{2m}|\psi(t)\rangle + \langle\vec{r}|V(\hat{\vec{r}},t)|\psi(t)\rangle, \tag{3.4.51}$$

   or writing the position representations explicitly,

$$i\hbar\frac{\partial}{\partial t}\psi(\vec{p},t) = \frac{\vec{p}^2}{2m}\psi(\vec{p},t) + V(i\hbar\vec{\nabla}_{\vec{p}},t)\psi(\vec{p},t). \tag{3.4.52}$$

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Question 3.4.6: The Schrödinger equation for the simple harmonic oscillator in one dimension in momentum representation is (A). $i\hbar\frac{\partial}{\partial t}\psi(x,t) = -\frac{\hbar^2}{2m}\frac{d^2\psi(x,t)}{dx^2} + \frac{1}{2}m\omega^2x^2\psi(x,t)$, (B). $i\hbar\frac{\partial}{\partial t}\psi(p,t) = -\frac{\hbar^2}{2m}\frac{d^2\psi(p,t)}{dx^2} + \frac{1}{2}m\omega^2x^2\psi(p,t)$, (C). $i\hbar\frac{\partial}{\partial t}\psi(p,t) = \frac{p^2}{2m}\psi(p,t) - \frac{\hbar^2}{2}m\omega^2\frac{d^2\psi(p,t)}{dp^2}$.

Question 3.4.7: Write down the wavefunction of the simple harmonic oscillator in momentum space using your knowledge of the wavefunctions in position space.

Exercise 3.4.6:

In problem 2.10.1, we investigated the Gaussian wave packet $\psi(x,t)$ in position representation (or position space), given $\psi(x,0) = Ae^{-\frac{x^2}{2a^2}}e^{ik_0x}$ with $A = \frac{1}{\sqrt{\sqrt{\pi}\,a}}$ when normalized. Now do the calculations below in the momentum representation (or momentum space) starting with $\phi(k,0) = \sqrt{\frac{a}{\sqrt{\pi}}}e^{-\frac{1}{2}a^2(k-k0)^2}$ that we obtained in 2.10.1(b).

1. Determine $\langle\hat{p}\rangle$ and $\langle\hat{p}^2\rangle$ at time t.
2. Determine $\langle\hat{x}\rangle$ and $\langle\hat{x}^2\rangle$ at time t.
3. Check that the uncertainty principle is satisfied with the minimum uncertainty saturated at $t = 0$.

3.5 Commuting operators

Consider a physical observable represented by Hermitian operator $\hat{A}$ for a system. Suppose we have solved the eigenvalue problem

$$\hat{A}|\psi\rangle = a|\psi\rangle; \tag{3.5.1}$$

i.e., we have found the eigenstates $|\psi\rangle$ and the corresponding real eigenvalues a of $\hat{A}$. Assume that the states are nondegenerate, with one eigenstate corresponding to one eigenvalue.

Next, consider a second physical observable represented by Hermitian operator $\hat{B}$. Suppose $\hat{A}$ and $\hat{B}$ commute; i.e.,

$$[\hat{A},\hat{B}] = 0. \tag{3.5.2}$$

We have

$$\hat{B}\hat{A}|\psi\rangle = \hat{B}\left(\hat{A}|\psi\rangle\right) = \hat{B}\left(a|\psi\rangle\right) = a\left(\hat{B}|\psi\rangle\right). \tag{3.5.3}$$

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On the other hand, because $\hat{B}\hat{A} = \hat{A}\hat{B}$, since $\hat{A}$ and $\hat{B}$ commute, we have

$$\hat{A}\hat{B}|\psi\rangle) = \hat{A}\left(\hat{B}|\psi\rangle\right) = \hat{B}\hat{A}|\psi\rangle = a\left(\hat{B}|\psi\rangle\right). \tag{3.5.4}$$

Therefore, $\hat{B}|\psi\rangle$ is an eigenstate of $\hat{A}$ with the same eigenvalue a as eigenstate $|\psi\rangle$. Because the states are nondegenerate and there is only one state with eigenvalue a, $\hat{B}|\psi\rangle$ and $|\psi\rangle$ can only differ by a constant, and we write

$$\hat{B}|\psi\rangle = b|\psi\rangle, \tag{3.5.5}$$

where b is a constant. In other words, $|\psi\rangle$ is a simultaneous eigenstate of $\hat{A}$ and $\hat{B}$. Because $|\psi\rangle$ is a simultaneous eigenstate of $\hat{A}$ and $\hat{B}$ with respective eigenvalues a and b, we rewrite it as

$$|\psi\rangle = |ab\rangle, \tag{3.5.6}$$

and we have

$$\hat{A}|ab\rangle = a|ab\rangle, \qquad \hat{B}|ab\rangle = b|ab\rangle. \tag{3.5.7}$$

Question 3.5.1: Suppose the simultaneous eigenstates of two commuting operators $\hat{A}$ and $\hat{B}$ are $|a_i b_i\rangle$. If a measurement of $\hat{A}$ is done and a_8 is obtained, then a following measurement of $\hat{B}$ in the same system produces (A). the ground state value of a, (B). 8, (C). b_8, (D). any value of b_i is possible.

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